Practice Questions

Candidate: Dunkan Gibson (dunkan.gibson)
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Score: 73% Passing Score: 80%

Question 1: ✔ Correct

You manage a network that uses switches. In the lobby of your building, there are three RJ45 ports connected to a switch.

You want to make sure that visitors cannot plug their computers into the free network jacks and connect to the network. But employees who plug into those same jacks should be able to connect to the network.

What feature should you configure?

- Bonding
- Mirroring
- ➡ Port authentication
- Spanning tree
- VLANs

EXPLANATION

Use port authentication to prevent unauthorized access through switch ports. Port authentication is provided by the 802.1x protocol and allows only authenticated devices to connect to the LAN through the switch. Authentication uses usernames and passwords, smart cards, or other authentication methods.

- When a device first connects, the port is set to an unauthorized state. Ports in unauthorized states can only be used for 802.1x authentication traffic.
- After the server authenticates the device or the user, the switch port is placed in an authorized state, and access to other LAN devices is allowed.

If you use a VLAN, you can assign each port to a VLAN. If the ports in the lobby were assigned to one VLAN, you could control the type of access through the switch for those ports, but could not modify the access based on user. Using a VLAN, both visitors and employees would have the same access through those ports.

Spanning tree is a protocol on a switch that allows the switch to maintain multiple paths between switches within a subnet. The spanning tree protocol runs on each switch and is used to select a single path between any two switches. Mirroring sends traffic from all switch ports to a switch port you designate as the mirrored port. Bonding allows multiple switch ports to be used at the same time to reach a specific destination.

REFERENCES

LabSim for Network Pro, Section 13.5.

Question 2: ✔ Correct

Which of the following is a platform independent authentication system that maintains a database of user accounts and passwords that centralizes the maintenance of those accounts?

- NAS
- ➡ RADIUS
- EAP
- RRAS

EXPLANATION

The Remote Authentication Dial-In User Service (RADIUS) is an authentication system that allows the centralization of remote user account management.

The Routing and Remote Access Service (RRAS) is a software component on a Windows Server system that provides remote access capabilities for users. A network access server (NAS) is a server or other system that acts as a gateway for remote user connections. The NAS passes authentication requests to the RADIUS server, which then checks the credentials of the user attempting to connect. NAS is also an acronym for network attached storage. Extensible authentication protocol (EAP) is an authentication protocol that supports the use of devices such as smart cards. It does not maintain a database of user accounts and passwords.

REFERENCES

LabSim for Network Pro, Section 13.5.

Question 3: ✔ Correct

Which of the following are methods for providing centralized authentication, authorization, and accounting for remote access? (Select two.)

- [ ] EAP
- [ ] 802.1x
- [x] RADIUS
- [x] TACACS+
- [ ] AAA
- [ ] PKI

EXPLANATION

Both RADIUS and TACACS+ are protocols used for centralized authentication, authorization, and accounting used with remote access. Remote access clients send authentication credentials to remote access servers. Remote access servers are configured as clients to the RADIUS or TACACS+ servers and forward the authentication credentials to the servers. The servers maintain a database of users and policies that control access for multiple remote access servers.

AAA stands for authentication, authorization, and accounting. AAA is a generic term that describes the functions performed by RADIUS/TACACS+ servers. A Public Key Infrastructure (PKI) is a system of certificate authorities that issue certificates. 802.1x is an authentication mechanism for controlling port access. 802.1x uses RADIUS/TACACS+ servers. EAP is an authentication protocol that allows the use of customized authentication methods.

REFERENCES

LabSim for Network Pro, Section 13.5.

Question 4: ✔ Correct

You want to implement an authentication method that uses public and private key pairs. Which authentication method should you use?

- PKI
- IPsec
- ➡ EAP
- MS-CHAP v2

EXPLANATION

Public and private key pairs are used by certificates for authentication and encryption. Extensible authentication protocol (EAP) allows the client and server to negotiate the characteristics of authentication. EAP is used to allow authentication using smart cards, biometrics (user physical characteristics), and certificate-based authentication.

MS-CHAP is Microsoft's proprietary method for remote access connections. MS-CHAP uses a three-way handshake (challenge/response) to perform authentication using a hashed form of a shared secret (password). A Public Key Infrastructure (PKI) is a system of certificate authorities that issue certificates, but is not a mechanism used for authentication. IPsec is a tunneling protocol used for VPN connections that provides encryption and a weak form of authentication using certificates, but is not used specifically for authentication.

REFERENCES

LabSim for Network Pro, Section 13.5.

Question 5: ✔ Correct

You have a web server that will be used for secure transactions for customers who access the website over the internet. The web server requires a certificate to support SSL.

Which method would you use to get a certificate for the server?

- ➡ (●) Obtain a certificate from a public PKI.
- ( ) Have the server generate its own certificate.
- ( ) Create your own internal PKI to issue certificates.
- ( ) Run a third-party tool to generate the certificate.

EXPLANATION

Computers must trust the CA that issues a certificate. For computers that are used on the internet and accessible to public users, obtain a certificate from a public CA such as VeriSign. By default, most computers trust well-known public CAs.

Use a private PKI to issue certificates to computers and users within your own organization. You configure computers to trust your own PKI, so certificates issued by your internal CAs are automatically trusted. A certificate generated by a server is called a self-signed certificate. A self-signed certificate provides no proof of identity because any other server can claim to be that server just by issuing itself a certificate.

REFERENCES

LabSim for Network Pro, Section 13.5.

Question 6: Incorrect

Which of the following is a feature of MS-CHAP v2 that is not included in CHAP?

- Certificate-based authentication
- Three-way handshake
- ~~Hashed shared secret~~
- Mutual authentication

EXPLANATION

MS-CHAP v2 allows for mutual authentication, where the server authenticates to the client.

Both CHAP and MS-CHAP use a three-way handshake process for authenticating users with usernames and passwords. The password (or shared secret) value is hashed, and the hash, not the shared secret, is sent for authentication.

REFERENCES

LabSim for Network Pro, Section 13.5.

Question 7: ✕ Incorrect

Which of the following actions typically involves the use of 802.1x authentication? (Select two.)

- [x] Enabling or disabling traffic on a port
- [ ] Authenticating VPN users through the internet
- [x] ~~Authenticating remote access clients~~
- [ ] Allowing authenticated users full access to the network
- [ ] Controlling access through a router

EXPLANATION

802.1x authentication is an authentication method used on a LAN to allow or deny access based on a port or connection to the network.

- The access point enables or disables traffic on the port based on the authentication status of the user.
- Authenticated users are allowed full access to the network while unauthenticated users only have access to the RADIUS server.

Remote access authentication is handled by remote access servers or a combination of remote access servers and a RADIUS server. VPN connections can be controlled by remote access servers or by a special device called a VPN concentrator.

REFERENCES

LabSim for Network Pro, Section 13.5.

Question 8: ✔ Correct

When using Kerberos authentication, which of the following terms is used to describe the token that verifies the user's identity to the target system?

- Voucher
- ➡ Ticket
- Hashkey
- Coupon

EXPLANATION

The tokens used in Kerberos authentication are known as tickets. Tickets perform a number of functions, including notifying the network service of the user who has been granted access and authenticating the identity of the person when they attempt to use that network service.

The terms coupon and voucher are not associated with Kerberos or any other commonly implemented network authentication system. The term hashkey is sometimes used to describe a value that has been derived from some piece of data if the value is used to access a service, but the term hashkey is not associated with Kerberos.

REFERENCES

LabSim for Network Pro, Section 13.5.

Question 9: ✔ Correct

You are a contractor that has agreed to implement a new remote access solution based on a Windows Server 2016 system for a client. The customer wants to purchase and install a smart card system to provide a high level of security to the implementation.

Which of the following authentication protocols are you most likely to recommend to the client?

- CHAP
- MS-CHAP
- PPP
- ➡ EAP

EXPLANATION

Of the protocols listed, only EAP provides support for smart card authentication.

The Microsoft Challenge Handshake Protocol (MS-CHAP) and the Challenge Handshake Protocol (CHAP) use a three-way handshake for authentication purposes. They do not support the use of smart cards. The point-to-point protocol (PPP) is a remote access protocol that uses usernames and passwords for authentication. It does not support the use of smart cards.

REFERENCES

LabSim for Network Pro, Section 13.5.

Question 10: ✔ Correct

Which of the following authentication methods uses tickets to provide single sign-on?

- PKI
- MS-CHAP
- ➡ Kerberos
- 802.1x

EXPLANATION

Kerberos grants tickets (also called security tokens) to authenticated users and to authorized resources. Kerberos uses the following components:

- An authentication server (AS) accepts and processes authentication requests.
- A service server (SS) is a server that provides or holds network resources.
- A ticket granting server (TGS) grants tickets that are valid for specific resources on specific servers.

802.1x is an authentication mechanism for controlling port access. 802.1x uses RADIUS/TACACS+ servers. MS-CHAP is Microsoft's proprietary method for remote access connections. MS-CHAP uses a three-way handshake (challenge/response) to perform authentication using a hashed form of a shared secret (password). A Public Key Infrastructure (PKI) is a system of certificate authorities that issue certificates, but is not a mechanism used for authentication.

REFERENCES

LabSim for Network Pro, Section 13.5.

Question 11: Incorrect

Which of the following applications typically use 802.1x authentication? (Select two.)

- [x] Controlling access through a wireless access point
- [ ] Controlling access through a router
- [x] Authenticating remote access clients
- [ ] Authenticating VPN users through the internet
- [ ] Controlling access through a switch

EXPLANATION

802.1x authentication is an authentication method used on a LAN to allow or deny access based on a port or connection to the network. 802.1x is used for port authentication on switches and authentication to wireless access points. 802.1x requires an authentication server for validating user credentials. This server is typically a RADIUS server.

Remote access authentication is handled by remote access servers or a combination of remote access servers and a RADIUS server for centralized authentication. VPN connections can be controlled by remote access servers or by a special device called a VPN concentrator.

REFERENCES

LabSim for Network Pro, Section 13.5.

Question 12: ✔ Correct

Which of the following is the most common form of authentication?

- Digital certificate on a smart card
- Fingerprint
- ➡ Password
- Photo ID

EXPLANATION

Passwords are the most common form of authentication. Most secure systems require only a username and password to provide users with access to the computing environment. Many forms of online intrusion attacks focus on stealing passwords. This makes using strong passwords very important. Without a strong password policy and properly trained users, the reliability of your security system is greatly diminished.

Photo ID, fingerprint, and digital certificate on a smart card are not the most common forms of authentication.

REFERENCES

LabSim for Network Pro, Section 13.5.

Question 13: ✔ Correct

Which of the following is the strongest form of multi-factor authentication?

- ➡ A password, a biometric scan, and a token device
- Two-factor authentication
- Two passwords
- A password and a biometric scan

EXPLANATION

A password, a biometric scan, and a token device together are the strongest form of multi-factor authentication listed here. Multifactor authentication is any combination of two or more of the same or different authentication factors. The three common authentication factor types are something you know (such as a password), something you have (such as a smart card or a token device), or something you are (such as a biometric quality like a fingerprint).

The other three options are all weaker forms of multi-factor authentication. A password and a biometric scan is a multi-factor authentication system, but it is also an example of two-factor authentication. Two-factor authentication is any combination of two or more different authentication factors. Two passwords is an example of multi-factor authentication, but since it uses two of the same type of factors, it is not a true two-factor authentication method.

REFERENCES

LabSim for Network Pro, Section 13.5.

Question 14: Incorrect

Which of the following is an example of three-factor authentication?

- ~~Smart card, digital certificate, PIN~~
- Token device, keystroke analysis, cognitive question
- Photo ID, smart card, fingerprint
- Pass phrase, palm scan, voice recognition

EXPLANATION

Three-factor authentication uses three items for authentication, one each from each of the authentication types:

- Type I (something you know, such as a password, PIN, pass phrase, or cognitive question)
- Type II (something you have, such as a smart card, token device, or photo ID)
- Type III (something you are, such as fingerprints, retina scans, voice recognition, or keyboard dynamics)

Of the examples listed, a token device (Type II), keystroke analysis (Type III), and a cognitive question (Type I) is the only three-factor authentication combination listed. The other options are examples of multi-factor authentication, where multiple authentication credentials, but not of three different types, are used.

REFERENCES

LabSim for Network Pro, Section 13.5.

Question 15: ✓ Correct

Match the authentication factor types on the left with the appropriate authentication factor on the right. Each authentication factor type can be used more than once.

PIN
- ✓ Something you know

Smart card
- ✓ Something you have

Password
- ✓ Something you know

Retina scan
- ✓ Something you are

Fingerprint scan
- ✓ Something you are

Hardware token
- ✓ Something you have

Voice recognition
- ✓ Something you are

Wi-Fi triangulation
- ✓ Somewhere you are

Typing behaviors
- ✓ Something you do

EXPLANATION

Something you know authentication requires you to provide a password or some other data. This is the weakest type of authentication. Examples of something you know authentication controls are:

- Passwords, codes, or IDs
- PINs
- Passphrases (long, sentence-length passwords)

Something you have (also called token-based authentication) is authentication based on something users have in their possession. Examples of something you have authentication controls are:

- Swipe cards
- Photo IDs
- Smart cards
- Hardware tokens

Something you are authentication uses a biometric system. A biometric system attempts to identify a person based on metrics or a mathematical representation of the subject's biological attribute. This is the most expensive and least accepted form of authentication, but is generally considered to be the most secure. Common attributes used for biometric systems are:

- Fingerprints
- Hand topology (side view) or geometry (top-down view)
- Palm scans
- Retina scans
- Iris scans
- Facial scans
- Voice recognition

Somewhere you are (also known as geolocation) is a supplementary authentication factor that uses physical location to verify a user's identity. Examples of implementations include:

- An account is locked unless the user has passed through the building's entrance using an ID card.
- If the user is within RFID range of the workstation, authentication requests are allowed.
- GPS or Wi-Fi triangulation location data is used to determine a device's location. If the user and the device are in a specified location, authentication requests are allowed. If not, the device is locked.

Something you do is a supplementary authentication factor that requires an action to verify a user's identity. Example implementations include:

- Analyzing a user's handwriting sample against a baseline sample before allowing authentication.
- Analyzing a user's typing behaviors against a baseline sample before allowing authentication.

REFERENCES

LabSim for Network Pro, Section 13.5.

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